KHADEEJA SHEREEF

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SUMMARY

Computer Science and Engineering graduate **(AI & ML)** with 9 months of hands-on R&D experience at ICFOSS, building OCR systems for Malayalam — a low-resource language. Progressed from a baseline pipeline to a modular, layout-aware system with measurable accuracy improvements at each stage. Experienced in deep learning, computer vision, data engineering, and dataset curation across the full ML pipeline — from raw data collection to model fine-tuning and evaluation.

EXPERIENCE

Analyst Intern | ICFOSS, Thiruvananthapuram, Kerala
July 2025 – April 2026

Project: **Platform Independent OCR Tool for Malayalam with Handwritten Text Recognition**

Phase 1 — Layout-Aware Printed Malayalam OCR Pipeline

- Built a baseline Tesseract-only OCR system for printed Malayalam documents, establishing a 40% text extraction accuracy benchmark across 2,000+ documents.
- Designed a layout-aware pipeline by integrating YOLOv11m for document region detection, achieving 93% layout detection accuracy across 3,000+ documents including complex newspaper layouts.
- Engineered sequential extraction logic where each detected layout region is processed in reading order, with image regions preserved as-is in the final output.
- Developed a document reconstruction module maintaining spatial fidelity of the source file using bounding box coordinates and aspect ratio, producing results in an editable format.
- Identified critical limitations of Tesseract in Malayalam text extraction and table recognition, forming the technical basis for Phase 2.

Phase 2 — DhritiOCR: Modular Malayalam OCR Pipeline (PaddleOCR-based)

- Architected DhritiOCR, a modular OCR pipeline built on PaddleOCR, replacing the Phase 1 system to address Tesseract's limitations in Malayalam text extraction and table handling.
- Fine-tuned a pretrained recognition model on Malayalam script data, achieving 92% character recognition accuracy — a significant improvement over the Phase 1 baseline.
- Integrated PP-DocLayoutV2 for document layout detection, routing each detected region (text, table, figure) to a specialised processor for accurate sequential extraction.
- Implemented table recognition using SLANeXt and RT-DETR models for structure recognition and cell detection, achieving 90% table extraction accuracy.

HTR Data Engineering — Handwritten Malayalam Dataset Preparation

- Designed and implemented a line segmentation pipeline from scratch using OpenCV for handwritten Malayalam documents, applying Otsu thresholding, horizontal morphological closing, and horizontal projection profiling to accurately isolate individual text lines.
- Engineered adaptive merging and padding logic to handle variable handwriting styles — preventing letter slicing across ascenders and descenders — and produced clean, training-ready line crops.
- Built and curated a structured handwritten Malayalam dataset, ensuring label quality and consistency to serve as a foundation for HTR model training.

Data Science and AI/ML Intern | Tech-By-Heart
July 2024 (1 week)

Python and Machine Learning Intern | Keltron
May 2023 (1 week)

EDUCATION

Bachelor of Technology in CSE (AI & ML)
Sree Buddha College of Engineering, 2021-2025 | CGPA:7.99
Higher Secondary
Govt Boys HSS, 2019-2021 | Percentage:98%

SKILLS

Programming Languages: Python, SQL
ML & Deep Learning: TensorFlow, PyTorch, Scikit-learn
Computer Vision & OCR: OpenCV, YOLOv11, PaddleOCR, Tesseract OCR, PP-DocLayoutV2
Data Engineering: Dataset curation, data cleaning, line segmentation, annotation pipelines
Libraries: NumPy, Pandas, Matplotlib, Seaborn
Database: MySQL

ACADEMIC PROJECTS

Deep Learning based Ophthalmic Disease Detection (VGG-19)

- Developed a deep learning model for ophthalmic disease detection using retinal fundus images, implementing VGG-19 with transfer learning and comparing performance against VGG-16.
- Deployed a web-based interface for real-time disease prediction and clinical use.

Machine learning based Student Performance Analysis System (SVM)

- Designed a predictive model using Support Vector Machine (SVM) trained on academic and demographic features to identify at-risk students.
- Developed a user interface to display prediction results and support early academic intervention.

CERTIFICATIONS

- **Introduction to Generative AI** — Google Cloud | May 2025
- **Python for Data Science** — IIT Madras, NPTEL | September 2024
- **Introduction to Data Analytics** — Simplilearn | June 2024
- **Introduction to Supervised and Unsupervised Machine Learning** — Simplilearn | December 2023